

Spatial Cognition Process Question Group

Positional Relationship (Obj.-Obj.)

Question: The rightmost large shelf in figure 2 is on the northwest side of the leftmost sofa in figure 1. In which direction is the simple desk in figure 2 relative to the leftmost sofa in figure 1? Options: A: east, B: south, C: north, D: west
Answer: D: west

Object Correspondence

Question: In Figure 2, there is a simple desk located at [0, 306, 146, 393]. Can you find this same physical object in Figure 1? If so, what are its coordinates in Figure 1?
Answer : Yes, the same simple desk is visible in Figure 1, and its coordinates are [499, 291, 639, 479].

Object Location

Question: In Figure2, please locate the simple desk and provide its bounding box coordinates.
Answer : The simple desk in Figure2 is located at [0, 306, 146, 393].

Camera Rotation (yaw)

Question: From Figure 2 to Figure 1, in which horizontal direction did the camera rotate, and what is the rotation angle?
Answer: The camera rotated left by approximately 107.65 degrees.

Camera-Object Yaw

Question: In Figure2, relative to the direction the camera is facing, what is the horizontal angular offset (yaw) of the simple desk? Is it to the left or right of the camera's center line?
Answer: In Figure2, the simple desk is positioned at a horizontal angle of 34.27 degrees to the left of the camera's central line of sight.

Level III: Contextual Reasoning

Counting

Positional Relationship (Cam.-Obj.)

Camera Motion

Size Comparison

Object Orientation

Virtual Perspective

Level II: Cross-View Understanding

Camera Rotation (pitch)

Camera Translation (forward)

Camera Translation (right)

Camera-Object 3D Position

Level I: Single-View Perception

Camera Pitch

Depth Estimation

Size Comparison

